Arpan Ghosh

Machine Learning · Data Science · AI Engineering

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SUMMARY

M.Tech CSE student at Tripura University. Builds and deploys end-to-end AI products: Attendly (face-and-voice attendance), NibbleContract AI (internship scam checker) and Movexa (real-time gym coach). IEEE-published coauthor in computer vision. Seeking remote ML, Data Science and AI Engineer internships.

EDUCATION

M.Tech (CSE), Tripura University — 2026 – 2028 · Pursuing

BE (CSE), Chandigarh University — 2021 – 2025 · CGPA 7.4

Class XII, WBCHSE · **Class X**, WBBSE — 93.4% · 87%

LANGUAGES

English
Hindi
Bengali (native)

SKILLS

ML & statistics: scikit-learn, XGBoost, SVM, regression, ensembles, clustering, PCA, SciPy

Deep learning: PyTorch, TensorFlow, ANN / FNN, CNN, RNN

Data & viz: SQL, pandas, NumPy, Matplotlib, Seaborn

Vision & audio: OpenCV, dlib, face & speaker embeddings, Resemblyzer, librosa

LLMs & agents: Agentic AI (Agno), RAG, LangChain, ChromaDB, Hugging Face, Groq & OpenAI APIs

Tools & infra: Python, Flask, Streamlit, Git, Docker, Render, Supabase, Jupyter

PROJECT EXPERIENCE

NibbleContract AI · Agentic Internship Scam Checker

Live demo · GitHub

Agentic AI, Agno, Python, Flask, Groq LLM, Pydantic, Tavily, DuckDuckGo, RDAP, pytest

- Built agentic AI with 3 parallel agents (reputation, company, funding) using web-search and domain-lookup tools; only source-grounded findings count, fixed rules set the score; 70+ pytest tests.
- Analyzes internship job posts before applying and offer letters after selection, giving an explainable 0–100 trust score, risk level, internship type (paid, unpaid, paid course in disguise) and evidence.
- Tested on 30 real documents, catching 90% of red flags with 93% fake-offer detection accuracy.

Attendly · AI Attendance with Face & Voice Recognition

Live demo · GitHub

Python, Streamlit, dlib ResNet embeddings, scikit-learn SVC, Resemblyzer, librosa, SQL

- Built class-wide attendance from one group photo: 128-d face embeddings plus an SVC classifier identify each student, and a distance threshold rejects unknown faces.
- Implemented voice roll-call by embedding similarity; teacher and student portals with secure logins.
- Achieved 95% face-match accuracy across 25 students, with 2% false-accept and 4% false-reject rates.

Movexa · AI Gym Coach & Live Form Tracker

Live demo · GitHub

Python, Pose estimation (33 landmarks), RAG, LLM API, Docker, WebRTC

- Built a personalised gym or home workout planner (sets, reps, calorie and protein targets) from body stats, goal and equipment, grounded by RAG over exercise-science guidance and adapted around injuries.
- Developed a live form coach: camera pose estimation measures joint angles, counts reps and sets, and gives on screen and spoken cues when form breaks; 54 exercises, 95% rep-count accuracy, ~30 ms per frame.

Machine Learning & Deep Learning Research

ML repo · DL repo

scikit-learn, XGBoost, PyTorch, TensorFlow / Keras, pandas, Matplotlib, Seaborn

- **Machine learning:** supervised and unsupervised learning on real datasets, from regression, SVM and ensembles to clustering, PCA and anomaly detection, compared through visualization.
- **Deep learning:** ANN / FNN, CNN and RNN models in PyTorch and TensorFlow for tabular, image and text tasks, plus a RAG pipeline with LangChain and ChromaDB.
- **Minor projects:** CreditWise Loan Approval · House Prices · Insurance Charges · Employee Turnover · Heart Disease · Titanic · Shop Smart · Thyroid Anomaly · Date Fruit · Power Plant · Handwritten Characters · ASL Sign Language

TRAINING & CERTIFICATIONS

Machine Learning, Internshala Trainings (6-week online training) — Aug 2023 · Certificate

AI/ML & Data Science, Apna College Prime 2.0 (ML, DL, SQL, web scraping, Gen AI, agentic AI, Docker) — 1 year · Completed

PUBLICATIONS

A login confirmation mechanism for meeting platforms with an automatic face acknowledgment system · Coauthor · IEEE CCICT 2024 · LBPH & Haar-cascade face recognition